

Isiri Amani Withanawasam

☎ +94 74 089 8813 | ✉ withanawasamia.21@uom.lk | 🔗 LinkedIn | 🐙 GitHub |
📍 Colombo, Sri Lanka

Interests

Computer Vision, Natural Language Processing, Federated Learning

Education

University of Moratuwa

B.Sc. in Electronic and Telecommunication Engineering; CGPA: 3.88/4.00

Moratuwa, Sri Lanka

September 2022 – June 2026

- **Relevant Coursework:** Linear Algebra, Calculus, Deep Learning for vision, Pattern Recognition, Image Processing and Machine Vision, Advances in Computer Vision

Devi Balika Vidyalaya

G.C.E Advanced Level: Combined Mathematics - A, Physics - A, Chemistry - A

Colombo 08, Sri Lanka

October 2020

- Z-Score - 2.3579 (Island Rank: 289/30,000 Top 0.96%)

Professional Experience

Pekoe Pvt Ltd - AI Engineer

July 2026 - Present

- Improving a Transformer-based tea auction price prediction model for production, focusing on accuracy enhancement and anomaly filtering.

A*STAR Institute of High Performance Computing, Singapore - Research Intern

July 2025 - December 2025
Supervisors : Dr . Vigneshwaran Subbaraju and Senior Scientist Vengateswaran Subramaniam

- Investigated data heterogeneity in Clustered Federated Learning by developing a multi-metric client clustering strategy that improves convergence and global model performance.

IronOne Technologies AI Labs, Sri Lanka - Internship

December 2024 - June 2025

Completed my six-month internship at IronOne Technologies - AI Labs, Sri Lanka, as a Machine Learning Engineering Intern. During my internship period I worked on two main projects.

- Cancer prediction using genomic data with interpretable ML visualizations, and YOLO-based mobile object detection using a custom dataset and selective layer freezing to improve precision.

Publications

PathKGen: Pathology Knowledge Assisted MLLM for Explainable Histopathology

Nadun Rajapaksha, Sandun Herath, Devnith Wijesinghe, **Isiri Amani Withanawasam**, Hasindri Watawana, Ranga Rodrigo

WACV 2027 Conference — *Under Review*.

Projects

PathKGen: Pathology Knowledge Assisted MLLM for Explainable Histopathology - Final Year Thesis Project

- Designing and implementing a vision language model for whole slide histopathology analysis by aligning giga-pixel WSI visual features with pathology-specific instructional text a curated knowledge graph.
- Fine-tuning a domain-specialized language model to generate credible, explainable diagnostic reasoning grounded in authoritative pathology references, and integrating it with a scalable WSI patching and aggregation pipeline.
- Curating a pathology-specific knowledge graph using UMLS database.
- **Technologies:** Python, PyTorch, VLMs, LLMs, Github Actions

Improving Clustered Federated Learning by Fusing Similarity Metrics - IHPC Project July 2025 - December 2025

- Building a clustered federated learning pipeline to improve client personalization and model generalization.
- Integrating Similarity Network Fusion (SNF) to combine multiple similarity metrics for improved client clustering. Extending and refactoring open-source implementations of EMDCFL, PACFL, and FlexCFL algorithms.
- Benchmarking clustering quality and model performance on CIFAR-10, Fashion-MNIST, and MNIST
- **Technologies:** Python, PyTorch, Clustered Federated Learning, Similarity Network Fusion (SNF), Github Actions, Tailscale

Genomic Cancer Prediction and Explainable AI - Internship Project December 2024 - June 2025

- Designed a ML pipeline for predicting variant pathogenicity using genomic features including AlphaMissense scores, GTEx tissue-specific expression (TPM), and epigenetic markers (DNA methylation, histone modifications).
- Developed custom Python scripts for variant-level data mapping and implemented standardized transcript selection using MANE and EMBL-EBI databases for biological accuracy.
- Enhanced model transparency and fairness using SHAP-based feature relevance via Microsoft RAI Dashboard and Uber Manifold for
- **Technologies:** Python, PyTorch, XGBoost, Docker, SHAP, RAI Dashboard

Computer Vision System for Industrial Bin Picking - Semester Project | February 2024 - June 2024

- Developed an industrial bin-picking vision pipeline using semantic segmentation to extract object grasp coordinates from RGB images.
- Implemented and benchmarked DeepLabV3, U-Net, SegNet, SAM, and FastSAM, optimizing accuracy-latency trade-offs for real-time and edge inference.
- Created and annotated a custom dataset, fine-tuned DeepLabV3 via transfer learning, and optimized inference using multithreading on desktop CPUs and Raspberry Pi 4B.
- **Technologies:** Python, PyTorch, OpenCV, Semantic Segmentation, Transfer Learning, Github Actions

Optimizing 12 Lead ECG Classification using VGGNet and ResNet - April 2024 - June 2024

- Developed and optimized deep learning pipelines using VGGNet and ResNet architectures to classify 12-lead ECGs from image representations derived from raw signals, achieving high accuracy in multi-class classification.
- Project recognized with accepted abstract at CINC Conference 2024 as part of the George B. Moody Physionet Challenge 2024 – Official Phase.
- **Technologies:** Python PyTorch, NumPy, OpenCV, Github actions

Technical Skills

Programming Languages: Python, C++, HTML/CSS
AI & Deep Learning: PyTorch, TensorFlow, Federated Learning, Langchain, Hugging Face
Computer Vision: OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib
Architectures: VLMs, Federated Learning Algorithms(IFCA, CFL, FLEX-CFL), YOLO, VGGNet
Medical Image Processing: ITK, VTK, ITK-Snap
Tools Git, GitHub, AWS, Docker
Soft Skills: Technical Writing, Compering, Problem-solving, Teamwork, Time Management

Achievements

Dean’s List Placement | Semesters 1, 2, 4, 5, 6, 7 and 8: Faculty of Engineering
ScholarX Mentee | Class of 2024
Abstract Accepted: CINC Conference 2024 | June 2024
9th China International College Students’ Internet+ Competition | June 2023
Spirit of Service Award | Rotaract Club of University of Moratuwa | May 2023

Certifications

Fundamentals of Deep Learning | NVIDIA | December 2025
Advanced Learning Algorithms | DeepLearning.ai via Coursera | February 2024
Convolutional Neural Networks | DeepLearning.ai via Coursera | February 2024

Volunteering

RoboticGen Academy | Instructor

July 2023 – May 2026

IEEE EMBS Student Branch Chapter: UoM

May 2023 – April 2026

- Brainstorm 2024: Editorial Lead
- Social Media Manager
- Co-Editor
- Councilor

Rotaract Club of University of Moratuwa

June 2022 – May 2023

- Career Fair Coordinator (Are You Ready 2023)
- Moderator
- MentHer OC Member

References

Dr. Ranga Rodrigo

Senior lecturer, Department of Electronic and
Telecommunication Engineering
University of Moratuwa
Email: ranga@uom.lk

Dr. Rukshani Liyanaarachchi

Senior lecturer, Department of Electronic and
Telecommunication Engineering
University of Moratuwa
Email: rukshanil@uom.lk